

Syllabus

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1. **Preliminaries to calculus :** Motivation for calculus. Sets and set operations. Basic propositional logic, quantifiers. Functions and relations. Types of functions. Open and closed sets. Countable and uncountable sets. Axiomatization of real numbers. Well ordering principle. Principle of recursive definition. Principle of mathematical induction. Basic inequalities. Triangle inequality. Some important inequalities. Archimedean property.
2. **Sequences and Series :** Definition of sequences and series. Supremum and infimum. Arithmetic progression. Geometric progression. Arithmetico-geometric progression. Tests of convergence. Radius of convergence. Trigonometrical series.
3. **Algebra and Basic Combinatorics :** Quadratic equations. Polynomials. Polynomial division. Partial fractions. Introduction to complex numbers. Basic counting. Pigeonhole principle. Combinations and permutations. Binomial theorem. Binomial coefficients and their summations, identities. Floor function and ceiling function. Their properties. More complex numbers.
4. **Limits :** Intuitive description of limit. Epsilon delta definition. Limit laws. Limit theorems. Continuity. Mean value theorems.
5. **Integral Calculus :** Partitions. Step functions. Area function. Calculating integrals.
6. **Differential Calculus :** Rate of change. Derivatives. Formulas of derivatives. Fundamental theorem of calculus. Extremization problems. Introduction to logarithms and exponential functions. Trigonometric functions and their calculus.
7. **Techniques of Integration :** All the standard techniques along with some extra techniques such as Feynman's rule, etc.

8. **Differential Equations** : Linear differential equations of first and second order. Some special ODEs and techniques.
9. **Classical Mechanics** : Kinematics. Newton's laws. Circular Motion. Central force definition. Rutherford scattering. Kepler's Laws. Gravity. Coulomb's law. Angular momentum. Torque. Principle axes. Euler's equations. Motion of a top. Coriolis force. Accelerated frames of reference.
10. **Trigonometry** : all the basics. Relation to complex numbers. Hyperbolic functions.
11. **Linear Algebra** : Linear equations. Matrices. Row operations. Reduced row echelon form. Invertibility of matrices. Determinants. Vector spaces. Linear transformations. Properties of determinants and matrices. Eigenvalues and eigenvectors. Eigenspaces. Cayley Hamilton theorem. Special types of matrices.
12. **Probability** : Conditional probability. Random variables. Discrete distributions. Continuous distributions. Binomial and Poisson distributions. Normal distribution. Bernoulli trials. Random variables. Law of large numbers.
13. **Special Relativity** : Postulates. Michelson Morley experiment. Basics of special relativity. Energy mass equivalence.
14. **Waves and Oscillations** : Simple harmonic motion. Forced oscillations. Resonance. Normal modes. Waves. Continuum limit. Elementary Fourier analysis. Longitudinal oscillations. Sound. Travelling waves.
15. **Electricity and Magnetism** : Some history. Electric charge. Electric potential. Gauss's Law. Laplace's equation. Stokes's theorem. Electric currents. Conductors. Magnetic field. Electromagnetic Induction. AC. Maxwell's equations. Electromagnetic waves. Electric and magnetic fields in matter.
16. **Optics and EM Waves** : Two and three dimensional waves. Spherical waves. Polarization. Interference and diffraction. Double slit experiment. Fermat's principle. Ray optics. Refraction and reflection laws.

17. Thermodynamics and Statistical Physics : Ideal gas equation. Heat. Temperature. Boltzmann distribution. Maxwell-Boltzmann distribution. Kinetic theory of gases. Pressure. Mean free path. Properties of gases. First law of thermodynamics. Second law of thermodynamics. Carnot's theorem and Carnot engines. Entropy. Equipartition theorem. Third law.